

Bi-Lipschitz invariance of box-counting dimension

Theorem: Let F be a nonempty bounded subset of $\mathbb{R}^n$ and let $f: F \rightarrow \mathbb{R}^m$ be a Lipschitz map. Then $\underline{\dim}_B f(F) \leq \underline{\dim}_B F$

and $\overline{\dim}_B f(F) \leq \overline{\dim}_B F$.

Proof: If $\{U_i\}_{i=1}^\infty$ is a δ -cover of F , then $\{f(U_i \cap F)\}_{i=1}^\infty$ is a $c\delta$ -cover of $f(F)$, where c is Lipschitz constant of F .
 $(\because \text{diam}(f(U_i \cap F)) \leq c \text{diam}(U_i \cap F) \leq c\delta)$
 $\therefore N_{c\delta}(f(F)) \leq N_\delta(F)$ for any $\delta > 0$.

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$$\Rightarrow \log N_{c\delta}(f(F)) \leq \log N_\delta(F)$$

$$\Rightarrow \frac{\log N_{c\delta}(f(F))}{-\log \delta} \leq \frac{\log N_\delta(F)}{-\log \delta}$$

$$\Rightarrow \frac{\log N_{c\delta}(f(F))}{-\log(c\delta) + \log c} \leq \frac{\log N_\delta(F)}{-\log \delta}$$

Taking $\liminf_{\delta \rightarrow 0^+}$, we get

$$\underline{\dim}_B f(F) \leq \underline{\dim}_B F$$

Similarly, taking $\limsup_{\delta \rightarrow 0^+}$, we get

$$\overline{\dim}_B f(F) \leq \overline{\dim}_B F$$

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Corollary: If f is a bi-Lipschitz map, then
 $\underline{\dim}_B f(F) = \underline{\dim}_B F$ & $\overline{\dim}_B f(F) = \overline{\dim}_B F$.

Proof: f & f^{-1} are both Lipschitz map.

$\therefore$ by the prev. theorem,

$$\underline{\dim}_B f(F) \leq \underline{\dim}_B F$$

$$\text{and } \underline{\dim}_B F = \underline{\dim}_B f^{-1}(f(F)) \leq \underline{\dim}_B f(F).$$

$$\Rightarrow \underline{\dim}_B f(F) = \underline{\dim}_B F.$$

Similarly for $\overline{\dim}_B$.

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Prop: Let $g: [0,1] \rightarrow \mathbb{R}$ be Lipschitz.

$$\text{Then } \underline{\dim}_B(\text{graph}(g)) = 1,$$

$$\text{where } \text{graph}(g) = \{(x, g(x)) : x \in [0,1]\}.$$

In particular, the above is true if $g \in C^1$.

Proof: Define $f: [0,1] \rightarrow \text{graph}(g)$ by

$$f(x) = (x, g(x)).$$

$$\begin{aligned} \text{Then } \|f(x) - f(y)\| &= \sqrt{(x-y)^2 + (g(x) - g(y))^2} \\ &\leq \sqrt{(x-y)^2 + c^2(x-y)^2} \\ &\leq \sqrt{1+c^2} |x-y|. \end{aligned}$$

$\Rightarrow f$ is Lipschitz.

$$\text{Also, } \|f(x) - f(y)\| \geq |x-y|$$

$\therefore f$ is bi-Lipschitz.

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$$\begin{aligned} \therefore \text{ By the bi-Lipschitz invariance, } \\ \underline{\dim}_B f([0,1]) &= \underline{\dim}_B [0,1] = 1 \\ \& \quad \overline{\dim}_B f([0,1]) &= \overline{\dim}_B [0,1] = 1 \\ \therefore \dim_B f([0,1]) &= 1 \\ \text{ i.e. } \dim_B \text{graph}(g) &= 1 \end{aligned}$$

Remark: Since the Hausdorff dimension is also bi-Lipschitz invariant,
 $\dim_H \text{graph}(g) = 1$ if g is Lipschitz.

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Relationship between the Hausdorff dimension and the box-counting dimensions:

Theorem: For any nonempty bounded subset F of $\mathbb{R}^n$,
 $\dim_H F \leq \underline{\dim}_B F \leq \overline{\dim}_B F$.

Proof: Let $s < \dim_H F$

We'll show that $\underline{\dim}_B F \geq s$

(This will imply $\underline{\dim}_B F \geq \dim_H F$).

Since $s < \dim_H F$, $\mathcal{H}^s(F) = \infty$

i.e. $\lim_{\delta \rightarrow 0^+} \mathcal{H}_\delta^s(F) = \infty$

$\Rightarrow \exists \delta_0 > 0$ s.t. $\mathcal{H}_\delta^s(F) > 1 \quad \forall 0 < \delta < \delta_0$.

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Let $N_\delta(F)$ be the least number of sets in any δ -cover of F .

Since $\mathcal{H}_\delta^s(F) > 1$,

$$N_\delta(F) \delta^s > 1$$

$$\Rightarrow \log N_\delta(F) + s \log \delta > 0$$

$$\Rightarrow \frac{\log N_\delta(F)}{-\log \delta} > s$$

Taking $\liminf$ as $\delta \rightarrow 0^+$, we get

$$\underline{\dim}_B F \geq s.$$

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Prop: Let F be any nonempty bounded subset of $\mathbb{R}^n$. Then $\underline{\dim}_B \bar{F} = \underline{\dim}_B F$

and $\overline{\dim}_B \bar{F} = \overline{\dim}_B F$, where $\bar{F}$ is the closure of F .

Proof: Exercise.

Cor: $\underline{\dim}_B F = 1$ if F is a dense subset of $[0,1]$.

In particular, $\underline{\dim}_B (\mathbb{Q} \cap [0,1]) = 1$.

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Prop: Let $\{\delta_k\}_{k=1}^{\infty}$ be a decreasing seq. of positive real numbers such $\delta_k \rightarrow 0$ as $k \rightarrow \infty$ and

$$\delta_{k+1} \geq c \delta_k \quad \text{for some } 0 < c < 1$$

(for example, $\delta_k = c^k$ for $0 < c < 1$)

Then $\underline{\dim}_B F = \lim_{k \rightarrow \infty} \frac{\log N_{\delta_k}(F)}{-\log \delta_k}$

and $\overline{\dim}_B F = \limsup_{k \rightarrow \infty} \frac{\log N_{\delta_k}(F)}{-\log \delta_k}$.

Proof: Clearly, $\underline{\dim}_B F = \liminf_{\delta \rightarrow 0^+} \frac{\log N_{\delta}(F)}{-\log \delta} \leq \liminf_{k \rightarrow \infty} \frac{\log N_{\delta_k}(F)}{-\log \delta_k}$.

We need to prove the opposite inequality.

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Let $0 < \delta < 1$. Choose $k \in \mathbb{N}$ s.t. $\delta_{k+1} \leq \delta < \delta_k$

$$\begin{aligned} \frac{\log N_{\delta}(F)}{-\log \delta} &\geq \frac{\log N_{\delta_k}(F)}{-\log \delta_{k+1}} \\ &= \frac{\log N_{\delta_k}(F)}{-\log \delta_k + \log\left(\frac{\delta_k}{\delta_{k+1}}\right)} \\ &\geq \frac{\log N_{\delta_k}(F)}{-\log \delta_k + \log\left(\frac{1}{c}\right)} \quad \left(\because \frac{\delta_k}{\delta_{k+1}} \leq \frac{1}{c}\right) \end{aligned}$$

Taking $\liminf$ as $k \rightarrow \infty$, we get

$$\underline{\dim}_B F \geq \liminf_{k \rightarrow \infty} \frac{\log N_{\delta_k}(F)}{-\log \delta_k}$$

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Exercise: Show that $\dim_{\mathbb{B}} F = \frac{\ln 2}{\ln 3}$ for the Cantor set F by using the previous proposition with $\delta_k = 3^{-k}$.

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